



Nitromethane

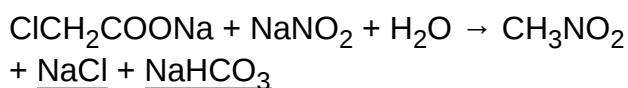
Nitromethane, sometimes shortened to simply "nitro", is an organic compound with the chemical formula CH_3NO_2 . It is the simplest organic nitro compound. It is a polar liquid commonly used as a solvent in a variety of industrial applications such as in extractions, as a reaction medium, and as a cleaning solvent. As an intermediate in organic synthesis, it is used widely in the manufacture of pesticides, explosives, fibers, and coatings.^[12] Nitromethane is used as a fuel additive in various motorsports and hobbies, e.g. Top Fuel drag racing and miniature internal combustion engines in radio control, control line and free flight model aircraft.

Preparation

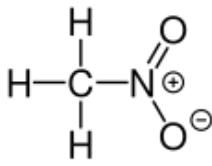
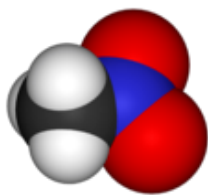
Nitromethane is produced industrially by combining propane and nitric acid in the gas phase at 350–450 °C (662–842 °F). This exothermic reaction produces the four industrially significant nitroalkanes: nitromethane, nitroethane, 1-nitropropane, and 2-nitropropane. The reaction involves free radicals, including the alkoxy radicals of the type $\text{CH}_3\text{CH}_2\text{CH}_2\text{O}\cdot$, which arise via homolysis of the corresponding nitrite ester. These alkoxy radicals are susceptible to C—C fragmentation reactions, which explains the formation of a mixture of products.^[12]

Laboratory methods

It can also be prepared by other methods that are of instructional value. The reaction of sodium chloroacetate with sodium nitrite in aqueous solution produces this compound:^[13]



Nitromethane

	
Names	
IUPAC name	Nitromethane
Preferred IUPAC name	Nitromethane ^[1]
Other names	Nitrocarbol
Identifiers	
CAS Number	75-52-5 (https://commonchemistry.cas.org/detail?cas_rn=75-52-5) ✓
3D model (JSmol)	Interactive image (https://chemapps.stolaf.edu/jmol/jmol.php?model=C%5BN%2B%5D%28%3DO%29%5BO-%5D)
ChEBI	CHEBI:77701 (https://www.ebi.ac.uk/chebi/searchId.do?chebiId=77701) ✗
ChEMBL	ChEMBL276924 (https://www.ebi.ac.uk/chembl/explore/compound/ChEMBL276924) ✓
ChemSpider	6135 (https://www.chemspider.com/Chemical-Structure.6135.html) ✓
ECHA InfoCard	100.000.797 (https://echa.europa.eu/substance-information/-/substanceinfo/100.000.797)
KEGG	C19275 (https://www.kegg.jp/entry/C19275) ✗

Uses

The dominant use of the nitromethane is as a precursor reagent. A major derivative is chloropicrin (CCl_3NO_2), a widely used pesticide. It condenses with formaldehyde (Henry reaction) to eventually give tris(hydroxymethyl)aminomethane ("tris"), a widely used buffer and ingredient in alkyd resins.^[12]

Solvent and stabilizer

The major application is as a stabilizer in chlorinated solvents. As an organic solvent, nitromethane has an unusual combination of properties: highly polar ($\epsilon_r = 36$ at 20 °C and $\mu = 3.5$ Debye) but aprotic and weakly basic. This combination makes it useful for dissolving positively charged, strongly electrophilic species. It is a solvent for acrylate monomers, such as cyanoacrylates (more commonly known as "super-glues").^[12]

Fuel

Although a minor application in terms of volume,^[12] nitromethane also is used as a fuel or fuel additive for sports and hobby. For some applications, it is mixed with methanol in racing cars, boats, and model engines.

Nitromethane is used as a fuel in motor racing, particularly drag racing, as well as for radio-controlled model power boats, cars, planes and helicopters. In this context, nitromethane is commonly referred to as "nitro fuel" or simply "nitro", and is the principal ingredient for fuel used in the "Top Fuel" category of drag racing.^[14]

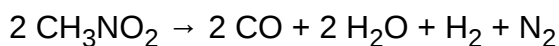
The oxygen content of nitromethane enables it to burn with much less atmospheric oxygen than conventional fuels.^[15] During nitromethane combustion, nitric oxide (NO) is one of the major emission products along with CO_2 and H_2O .^[16] Nitric oxide contributes to air pollution, acid rain, and ozone layer depletion. Recent (2020) studies^[17] suggest the correct stoichiometric equation for the burning of nitromethane is:

PubChem CID	6375 (https://pubchem.ncbi.nlm.nih.gov/compound/6375)
RTECS number	PA9800000
UNII	RU5WG8C3F4 (https://precision.fda.gov/uniisearch/srs/unii/RU5WG8C3F4) ✓
CompTox Dashboard (EPA)	DTXSID2020977 (https://comptox.epa.gov/dashboard/chemical/details/DTXSID2020977)
InChI	InChI=1S/CH3NO2/c1-2(3)4/h1H3 ✓ Key: LYGJENNIWJXYER-UHFFFAOYSA-N ✓
InChI=1/CH3NO2/c1-2(3)4/h1H3	Key: LYGJENNIWJXYER-UHFFFAOYAW
SMILES	<chem>C[N+](=O)[O-]</chem>
Properties	
Chemical formula	CH_3NO_2
Molar mass	61.04 g/mol
Appearance	colorless, oily liquid ^[2]
Odor	Light, fruity ^[2]
Density	1.1371 g/cm ³ (20 °C) ^[3]
Melting point	−28.7 °C (−19.7 °F; 244.5 K) ^[3]
Boiling point	101.2 °C (214.2 °F; 374.3 K) ^[3]
Critical point (T, P)	588 K, 6.0 MPa ^[4]
Solubility in water	ca. 10 g/100 mL
Solubility	miscible in <u>diethyl ether</u> , <u>acetone</u> , <u>ethanol</u> , <u>methanol</u> ^[3]
Vapor pressure	28 mmHg (20 °C) ^[2]
Acidity (pK _a)	10.21 (H ₂ O) ^[5] 17.2 (DMSO) ^[6]
Magnetic susceptibility (χ)	−21.0·10 ^{−6} cm ³ /mol ^[7]
Thermal conductivity	0.204 W/(m·K) at 25 °C ^[8]
Refractive index (n _D)	1.3817 (20 °C) ^[3]
Viscosity	0.63 cP at 25 °C ^[8]



The amount of air required to burn 1 kg (2.2 lb) of gasoline is 14.7 kg (32 lb), but only 1.7 kg (3.7 lb) of air is required for 1 kg of nitromethane. Since an engine's cylinder can only contain a limited amount of air on each stroke, 8.6 times as much nitromethane as gasoline can be burned in one stroke. Nitromethane, however, has a lower specific energy: gasoline provides about 42–44 MJ/kg, whereas nitromethane provides only 11.3 MJ/kg. This analysis indicates that nitromethane generates about 2.3 times the power of gasoline when combined with a given amount of oxygen.

Nitromethane can also be used as a monopropellant, i.e., a propellant that decomposes to release energy without added oxygen. It was first tested as rocket monopropellant in 1930s by Luigi Crocco from Italian Rocket Society.^{[18][19]} There is a renewed interest in nitromethane as safer replacement of hydrazine monopropellant.^[20] The following equation describes this process:



Nitromethane has a laminar combustion velocity of approximately 0.5 m/s, somewhat higher than gasoline, thus making it suitable for high-speed engines. It also has a somewhat higher flame temperature of about 2,400 °C (4,350 °F). The high heat of vaporization of 0.56 MJ/kg together with the high fuel flow provides significant cooling of the incoming charge (about twice that of methanol), resulting in reasonably low temperatures.

Nitromethane is usually used with rich air–fuel mixtures because it provides power even in the absence of atmospheric oxygen. When rich air–fuel mixtures are used, hydrogen and carbon monoxide are two of the combustion products. These gases often ignite, sometimes spectacularly, as the normally very rich mixtures of the still burning fuel exits the exhaust ports. Very rich mixtures are necessary to reduce the temperature of combustion chamber hot parts in order to control pre-ignition and subsequent detonation. Operational details depend on the particular mixture and engine characteristics.

Dipole moment	3.46 ^[9]
Explosive data	
Shock sensitivity	Low
Friction sensitivity	Low
Detonation velocity	6400 m/s
Thermochemistry^[10]	
Heat capacity (C)	106.6 J/(mol·K)
Std molar entropy ($S^\ominus_{298}$)	171.8 J/(mol·K)
Std enthalpy of formation ($\Delta_f H^\ominus_{298}$)	−112.6 kJ/mol
Gibbs free energy ($\Delta_f G^\ominus$)	−14.4 kJ/mol
Hazards	
Occupational safety and health (OHS/OSH):	
Main hazards	Flammable, health hazard
GHS labelling:	
Pictograms	
Signal word	Danger
Hazard statements	H203, H226, H301, H331, H351
Precautionary statements	P210, P261, P280, P304+P340, P312, P370+P378, P403+P233
NFPA 704 (fire diamond)	
Flash point	35 ^[9] °C (95 °F; 308 K)
Autoignition temperature	418 ^[9] °C (784 °F; 691 K)
Explosive limits	7–22% ^[9]
Threshold limit value (TLV)	20 ppm ^[9]
Lethal dose or concentration (LD, LC):	
LD ₅₀ (median dose)	940 mg/kg (oral, rat) 950 mg/kg (oral, mouse) ^[11]
LD _{Lo} (lowest published)	750 mg/kg (rabbit, oral) 125 mg/kg (dog, oral) ^[11]

A small amount of hydrazine blended in nitromethane can increase the power output even further. With nitromethane, hydrazine forms an explosive salt that is again a monopropellant. This unstable mixture poses a severe safety hazard. The National Hot Rod Association and Academy of Model Aeronautics do not permit its use in competitions.^[21]

In model aircraft and car glow fuel, the primary ingredient is generally methanol with some nitromethane (0% to 65%, but rarely over 30%, and 10–20% lubricants (usually castor oil and/or synthetic oil)). Even moderate amounts of nitromethane tend to increase the power created by the engine (as the limiting factor is often the air intake), making the engine easier to tune (adjust for the proper air/fuel ratio).

Former uses

It formerly was used in the explosives industry as a component in a binary explosive formulation with ammonium nitrate and in shaped charges, and it was used as a chemical stabilizer to prevent decomposition of various halogenated hydrocarbons.^[22]

Other

It can be used as an explosive, when gelled with several percent of gelling agent. This type of mixture is called PLX. Other mixtures include ANNM and ANNMAI – explosive mixtures of ammonium nitrate, nitromethane and aluminium powder.

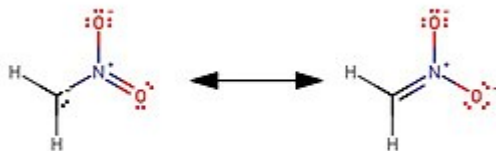
A YouTuber posted a video that demonstrated that burning nitromethane will give off a very unusually colored flame.^[23] The flame actually appears to be black and white. He has used methanol to start the fire in the mentioned video.

Reactions

Acid-base properties

Nitromethane is a relatively acidic carbon acid. It has a pK_a of 17.2 in DMSO solution. This value indicates an aqueous pK_a of about 11.^[24] It is so acidic because the anion admits an alternate, stabilizing resonance structure:

LC _{Lo} (lowest published)	7087 ppm (mouse, 2 h) 1000 ppm (monkey) 2500 ppm (rabbit, 12 h) 5000 ppm (rabbit, 6 h) ^[11]
NIOSH (US health exposure limits):	
PEL (Permissible)	TWA 100 ppm (250 mg/m ³) ^[2]
REL (Recommended)	none ^[2]
IDLH (Immediate danger)	750 ppm ^[2]
Related compounds	
Related nitro compounds	<u>nitroethane</u>
Related compounds	<u>methyl nitrite</u> <u>methyl nitrate</u>
Supplementary data page	
<u>Nitromethane (data page)</u>	
Except where otherwise noted, data are given for materials in their <u>standard state</u> (at 25 °C [77 °F], 100 kPa).	
✗ <u>verify</u> (what is ✓✗ ?) <u>Infobox references</u>	



The acid deprotonates only slowly. Protonation of the conjugate base O_2NCH_2^- , which is nearly isosteric with nitrate, occurs initially at oxygen.^[25]

Organic reactions

In organic synthesis nitromethane is employed as a one carbon building block.^{[26][27]} Its acidity allows it to undergo deprotonation, enabling condensation reactions analogous to those of carbonyl compounds. Thus, under base catalysis, nitromethane adds to aldehydes in 1,2-addition in the nitroaldol reaction. Some important derivatives include the pesticides chloropicrin (Cl_3CNO_2), beta-nitrostyrene, and tris(hydroxymethyl)nitromethane, $((\text{HOCH}_2)_3\text{CNO}_2)$. Reduction of the latter gives tris(hydroxymethyl)aminomethane, $(\text{HOCH}_2)_3\text{CNH}_2$, better known as tris, a widely used buffer. In more specialized organic synthesis, nitromethane serves as a Michael donor, adding to α,β -unsaturated carbonyl compounds via 1,4-addition in the Michael reaction.

Purification

Nitromethane is a popular solvent in organic and electroanalytical chemistry. It can be purified by cooling below its freezing point, washing the solid with cold diethyl ether, followed by distillation.^[28]

Safety

Nitromethane has a modest acute toxicity. LD50 (oral, rats) is 1210 ± 322 mg/kg.^[12]

Nitromethane is "reasonably anticipated to be a human carcinogen" according to a U.S. government report.^[29]

Explosive properties

Nitromethane was not known to be a high explosive until a railroad tank car loaded with it exploded on June 1, 1958.^[30] After much testing, it was realized that nitromethane was a more energetic high explosive than TNT, although TNT has a higher velocity of detonation (VoD) and brisance. Both of these explosives are oxygen-poor, and some benefits are gained from mixing with an oxidizer, such as ammonium nitrate. Pure nitromethane is an insensitive explosive with a VoD of approximately 6,400 m/s (21,000 ft/s), but even so inhibitors may be used to reduce the hazards. The tank car explosion was speculated to be due to adiabatic compression, a hazard common to all liquid explosives. This is when small entrained air bubbles compress and superheat with rapid rises in pressure. It was thought that an operator rapidly snapped shut a valve creating a "hammer-lock" pressure surge.

If mixed with ammonium nitrate, which is used as an oxidizer, it forms an explosive mixture known as ANNM.

Nitromethane is used as a model explosive, along with TNT. It has several advantages as a model explosive over TNT, namely its uniform density and lack of solid post-detonation species that complicate the determination of equation of state and further calculations.

Nitromethane reacts with solutions of sodium hydroxide or methoxide in alcohol to produce an insoluble salt of nitromethane. This substance is a sensitive explosive which reverts to nitromethane under acidic conditions and decomposes in water to form another explosive compound, sodium methazonate, which has a reddish-brown color:



Nitromethane's reaction with solid sodium hydroxide is hypergolic.

See also

- Top Fuel
- Adiabatic flame temperature, a thermodynamic calculation of the flame temperature of nitromethane
- Dinitromethane
- Model engine
- Trinitromethane
- Tetranitromethane
- RE factor

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Further reading

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External links

- WebBook page for nitromethane (<http://webbook.nist.gov/cgi/cbook.cgi?ID=C75525>)
- History of Nitromethane (<http://www.dragtimes.com/Nitromethane-Drag-Racing-Top-Fuel-So-up-of-Choice.html>)

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